

Fertility Risk and the Life-Cycle

Sekyu Choi
Universitat Autònoma de Barcelona*

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Abstract

Two out of every four pregnancies is deemed unwanted by women in the United States. Around half of those are aborted. Given that children represent important expenses for households in terms of money and time, the presence of this prevalent (and "unremedied") unwantedness speaks about sources of non-monetary shocks that can determine aggregate differences in life-cycle consumption, savings and inequality across individuals.

In this paper I present a life-cycle model of fertility decisions in an environment where agents are subject to uninsurable labor income shocks and are unable to control perfectly the timing of pregnancies, the latter being a novel feature in life-cycle economies with incomplete markets. My model also takes into account that fertility choices are intertwined with both labor and marriage market conditions which are heterogeneous across individuals and determine how the trade-off between increasing family size and the status quo looks like for females during the life-cycle.

I estimate the model using data from the 1995 National Survey of Family Growth (NSFG), the Panel Study of Income Dynamics (PSID) and the Current Population Survey (CPS). By matching aggregate education-specific fertility and abortion profiles, I find that fertility risk (the degree to which individuals fail to obtain their desired outcomes) differs markedly by education groups and is an important ingredient in explaining the differences observed in the data. This result shows that fertility risk is a source of lifetime inequality -since it is more prevalent among lower educated individuals- and highlights the importance of incorporating endogenous and imperfect fertility decisions into life-cycle models when the studied environments exhibit incomplete insurance.

Keywords: Stochastic fertility, Life-cycle model, Heterogeneous agents, Birth control, Educational attainment

JEL Classification: D31,D91,E21,J13,J31

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1 Introduction

Around half the pregnancies in the United States are unwanted and half of those end up in an abortion (Henshaw (1998)). Since the arrival of children into a household represents important costs in terms of resources and time (hence, forgone wages), the seemingly imprecision with which individuals carry out their fertility plans can be identified as an important source of lifetime inequality.

In this paper I pose a life-cycle, consumption-savings framework with uninsurable earnings risks and rich demographics. The structure is similar to the models in Hong and Ríos-Rull (2007) and Hong (2008), which I extend by considering fertility as an endogenous choice, subject to idiosyncratic shocks (i.e., fertility risk). I then estimate the model using data from the 1995 National Survey of Family Growth (NSFG), the Panel Study of Income Dynamics (PSID) and the Current Population Survey (CPS) using the simulated method of moments, where I choose education-specific fertility and abortion profiles as data moments to be matched by the model.

Results from the quantitative exercise show that differential fertility risk is important when trying to account for differences in the timing and the number of births across educational groups as well as unwanted and abortion rates. This contrasts with previous literature like Mincer (1963), Becker (1965), Galor and Weil (1996) and Greenwood, Seshadri, and Vandenbroucke (2005), among others, where the higher opportunity cost of child bearing incurred by higher skilled individuals (having and rearing children is a time consuming activity) delivers a lower demand for children and a higher demand for abortions in case of birth control mishaps. This last mechanism is present in this paper but ends up playing a minor role quantitatively.

My setup is different given the presence of incomplete markets, heterogeneous income profiles and a realistic marriage market: under imperfect insurance mechanisms, non-labor income (return from savings) is endogenous and higher for more skilled/lucky earners. This creates an income effect that can overpower the opportunity cost of having children. In other words, higher educated individuals with higher wages, can insure better against the costs of child rearing. This effect is reinforced if we account for marriage and positive assortative matching: females with more education tend to marry more educated males, who have higher wages. Just like own savings, male wages can act as an insurance against child expenses and the consumption cost of not working due to child rearing.¹

The result above depends on the imposed preference structure, specifically the degree of substitutability/complementarity between consumption and "children services" in the utility function.² This determines the optimal timing and spacing of children and thus, informs about the relevance of fertility

¹In a related paper, Stetsenko (2009) studies a model with endogenous female labor force participation and fertility and finds that the interaction between female and male life-cycle earnings profiles is important when accounting for increasing female labor force participation in the US.

²An informative discussion is found in Jones and Schoonbroodt (2009), who show that a low intertemporal elasticity of substitution is needed for a Barro-Becker model to explain the demographic transition in the face of decreasing child mortality and increasing real productivity.

risk in the face of the timing of abortions as observed in the data. Thus, in my approach I identify these preference parameters and the amount of fertility risk, using the differential timing of pregnancies, (un)wantedness and abortion rates across different education groups.

This paper contributes to the literature that studies the sources of lifetime inequality using models with heterogenous households and incomplete markets. As pointed out by Heathcote, Storesletten, and Violante (2009), family composition shocks are an important source of both uninsurable economic risks but also insurance.³ In this paper I identify fertility shocks as a source of risk and heterogeneity, which suggests a reconsideration of models used to study female labor participation in which fertility is treated as exogenous⁴, and also of models in which fertility and labor market decisions are both endogenous but where fertility risk is not considered.

My model is closely related to Conesa (2000) and Sommer (2009): they both study life-cycle fertility decisions in environments with uninsurable labor income risk. I extend this framework by considering a richer demographic structure and the presence of fertility risk. On the other hand, the main mechanism in this paper is related to the findings in Rosenzweig and Schultz (1989) who show that more educated individuals are more efficient using different birth control methods.

My approach borrows insights from the empirical microeconomic literature that studies life-cycle fertility⁵ using structural and dynamic models of fertility choice. From that literature, my paper relates the most to Wolpin (1984) and Hotz and Miller (1993) who acknowledge the importance of the stochastic nature of fertility. Wolpin analyzes how child mortality risk shapes fertility choices using Malaysian data; Hotz and Miller estimate birth control method choices by females in a life-cycle framework. However, my approach differs in terms of assumptions regarding capital markets and preference heterogeneity: I assume imperfect capital markets in the sense that agents can save but not borrow against their future earnings while the above mentioned literature assumes away the existence of capital markets; also, I impose the same utility function for all agents, downplaying the role of unobserved heterogeneity in preferences to account for the data.

The structure of the paper is as follows: In the next section I describe my data sources and the main stylized facts I want to explain. In section 3, I describe my quantitative model. Sections 4 and 5 describe the functional forms used in the model and the specific estimation method to obtain model parameters. I show the estimation results and some quantitative experiments in section 6. The final section concludes.

³See for example, Cubeddu and Ríos-Rull (2003) and Kaplan (2009)

⁴See for example, Jones, Manuelli, and McGrattam (2003), Olivetti (2006) and Attanasio, Low, and Sánchez-Marcos (2008), among others

⁵See Hotz, Klerman, and Willis (1997) for a survey

2 The Facts

I use information from the National Survey of Family Growth (NSFG) to put forward a set of facts on U.S. fertility. The NSFG is compiled by the National Center for Health Statistics (NCHS) and gathers information on family life, fertility, use of birth control and other health related questions. I use the survey for the year 1995, which comprises around ten thousand women between the ages of 15 and 44.

For every survey participant, the NSFG collects retrospective information on usage of birth control methods, on a monthly basis for up to 5 years. Participants also answer questions on wantedness and timing of births and pregnancy outcomes for all pregnancies conceived during that 5 year period. The survey also contains information on educational attainment, marital status and other background information.

I present age specific fertility rates in figure 1 and age specific abortion rates in figure 2.

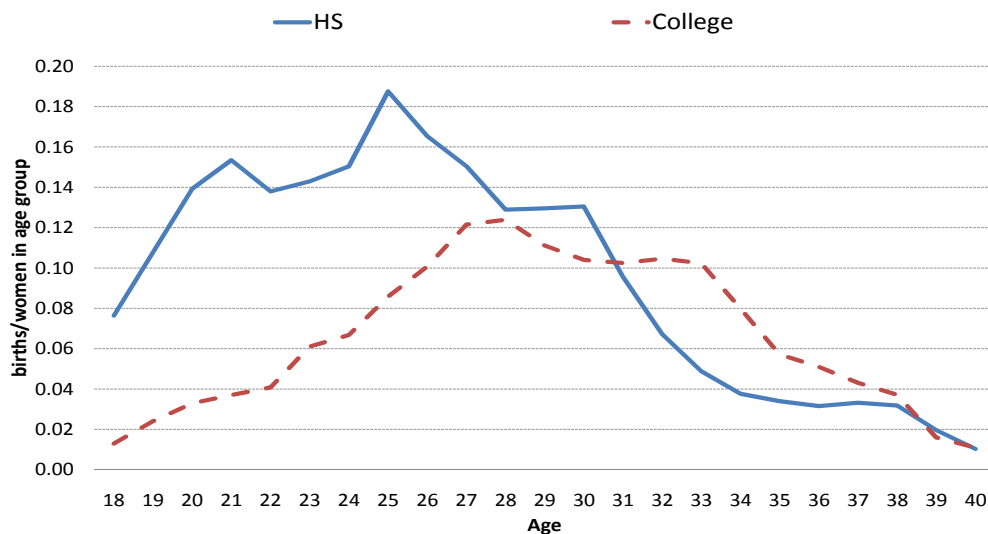


Figure 1: Age-specific fertility rates by education of the mother (NSFG 1995)

Both graphs present information on pregnancies occurring between 1994 and 1995, for each particular education-age group, i.e., I'm focusing on the cross-sectional dimension of the data.⁶ Age specific fertility rates are defined by the ratio between the number of pregnancies in the specific education-age group and the total number of women in that group. Abortion rates are the number of abortions divided by the total number of women in each group. I divide groups according to educational attainment as follows: High School (those without any post-secondary education) and College (those with at least some post-secondary education). In both figures, I show smoothed statistics (moving averages of 3 years).

⁶Hence, I'm assuming NO cohort effects in fertility rates. A more complete discussion is in the Appendix.

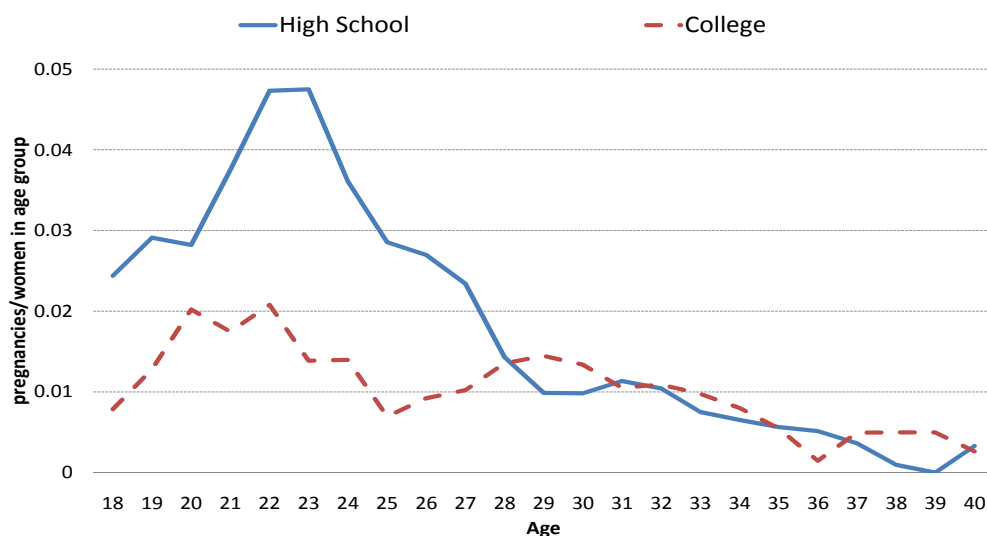


Figure 2: Age-specific abortion rates by education of the mother (NSFG 1995)

The following is a list of stylized facts from the data:

1. The education fertility gap: the high school group has a total fertility rate (TFR)⁷ of 2.2, while for the college group, it is 1.5 .
2. Timing of births: high school females start having children earlier than their college counterparts. According to figure 1, the age with the highest fertility rate is 25 for the High school group and 28 for the college one.⁸
3. Failure in fertility plans: the number of aborted pregnancies is higher for the high school group. The abortion rate for high school educated females is approximately 18 per 1000 women, while the number for college educated females is 11

All but the last fact have been well documented in recent economic literature. Since income of more educated individuals is higher, fact 1 above can be restated as the well known negative income-fertility relationship.⁹ The differential timing of births is documented and studied by Caucutt, Gunner, and Knowles (2002), who argue that returns to experience as well as marriage markets play an important role in explaining delay in childbirth.

⁷Total fertility rates are the sum of of the age specific fertility rates and represent a cross-sectional measure of aggregate fertility

⁸A similar statistic is average age of first birth (22.7 and 26.2 for high school and college respectively)

⁹This observation goes back to Becker (1960). Jones and Tertilt (2008) study Census data and find that this negative relationship is robust across time and different definitions of income

The last fact shows that failure rates are more acute for the high school group.¹⁰ Figure 3 presents a similar perspective: we see that the proportion of non-aborted pregnancies that were claimed as unwanted is higher for the high school group. For the entire sample, the average is around 45-50%, which corresponds roughly to the findings in Henshaw (1998), i.e., half of unwanted pregnancies ends up as an abortion. However, this figure also shows that the timing of unwanted pregnancies is concentrated both at the beginning and at the end of the years considered, a mirror image of the abortion rates in figure 2.

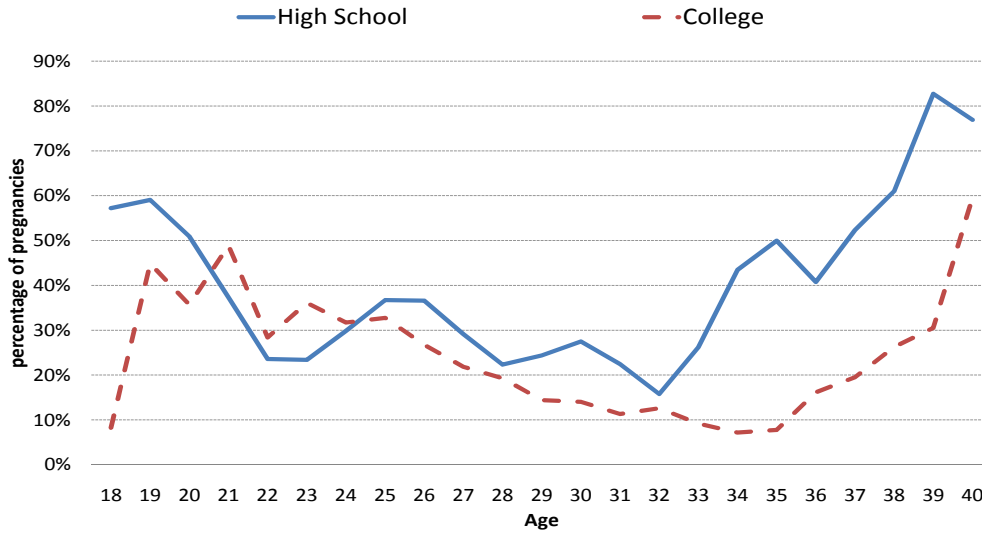


Figure 3: Unwanted minus aborted pregnancies by age and education of the mother (NSFG 1995)

3 A Quantitative Model

The model environment is an economy populated by agents of different gender (males and females) and education level (high and low)¹¹ who live finite lives and face three types of exogenous and idiosyncratic shocks: to their life (survival shocks), to their household type (marital transition shocks) and to their earnings (shocks to the value of their market rewards). All agents derive utility from consumption and from the presence of children in the household. Agents supply labor inelastically to the market before

¹⁰In terms of accuracy of this data, Fu, Darroch, Henshaw, and Kolb (1998) claim that the introduction of computer assisted interviews in the NSFG for the year 1995 helped in reducing underreporting of abortions and unplanned pregnancies. Nevertheless, their study shows (by comparing implied abortion rates from the NSFG to data from abortion providers in the U.S.) that non reported abortion cases are still present and are higher for lower income groups (the high school group). Hence, differences in failure rates by educational groups are likely to be more pronounced if the data did not present any missed observations in the form of misreported abortions.

¹¹I abstract from the education decisions in order to simplify the analysis and given that most fertility outcomes occur after the age of 22.

retirement and every period they decide how much to consume and save for the future; they cannot borrow.

During the first part of their life-cycle, female agents are fertile (can conceive children) and decide on contraceptive efforts period by period. This effort influences imperfectly the probability of conception. Unwanted pregnancies can be aborted; both contraceptive effort and abortions come at a utility cost. After a birth, female agents must spend some time at home rearing their children. Male agents are not affected by this requirement.

State space. Let z be the state space that defines an agent in this economy. Throughout the discussion, I focus on the female's point of view:

$$z = \{e, e^*, i, k, m, \epsilon, \epsilon^*, i^*, a\} \quad (1)$$

Asterisks represent values for spouses (when applicable). Age is indexed by $i = \{i_0, \dots, I\}$, $k = \{1, 2, \dots, K\}$ represents the number of children living in the household (not the same as parity), $m = \{1, 2, 3\}$ is the type of household (1 = single, 2 = married, 3 = widowed/divorced¹²), $e \in \{\underline{e}, \bar{e}\}$ represents the education type of the agent (low, high), ϵ is the value of the multiplicative shock to labor earnings and a is the amount of real assets in the household.

For ease of exposition, in some sections of the paper I use the following partition of the state space $\tilde{z} = \{m, e, e^*, \epsilon, \epsilon^*, i^*\}$ so that $z = \{i, a, k\} \times \tilde{z}$.

The Life-cycle proper. All agents start life at age i_0 (first year of adulthood) being one of two educational types: low ($\underline{e}$) or high ($\bar{e}$). This type doesn't change and can be considered as a decision taken before the events in the model. Agents can also start life as married or single and with or without children.

The maximum lifespan for all agents is of I years. Survival from age i to $i + 1$ is subject to state dependent mortality risk, i.e., the probability of surviving an additional year depends on the gender and the educational type of the agent. I denote this probability as $\delta_{i,e}$. The probability for males is δ_{i^*,e^*}^* .

With regard to labor markets, agents work until they reach age i_r . The retirement age is common for males and females. Female agents also make fertility decisions from i_0 to i_f , the last fertile age. This cut-off for the fertile period is common and known to all female agents.

Fertility and children. During their fertile years, females choose effort to determine the probability of a pregnancy. I denote this effort as $x \in \mathbb{R}$, which translates into a probability $\pi(x|i, m, e) \in (0, 1)$ of *no* conception. This stochastic production function of no pregnancies depends on the age of the

¹²Features of widowed vs. divorced households are unified in a single state, since their distinctions in the data are not significant

female agent (to capture biological constraints on women's reproductive systems), her marital status (since conception opportunities might differ if a mate is present or not) and her education. Evidence of this last point is in Rosenzweig and Schultz (1989), who estimate differential effectiveness rates of contraceptive use by educational attainment. The exertion of this effort comes at a utility cost of $C(x)$.

With complementary probability $(1 - \pi)$, a pregnancy occurs. If the pregnancy falls into the category of "unplanned/unwanted" (i.e., a positive amount of contraceptive effort was exerted), agents have the opportunity of getting an abortion at a utility cost κ_e . This cost depends on the educational level of the agent. If the pregnancy is intended (i.e., $x < 0$) the agent keeps the child and the household increases its size by one.¹³

I make the assumption that children are attached to female agents. I don't keep track of the age nor the sex of children in the household for reasons of computational burden. Instead, households face a constant hazard rate for the permanence of children in the household. I denote this hazard by s_k , which means that on average, children spend $1/s_k$ periods attached to their mothers.¹⁴ Finally, no children can stay in the household after retirement of the mother.

Marital states. The transition through different marital status is stochastic and exogenous. The probability of going from m to m' (conditional on both spouses being alive, in case of agents being married) is given by $\Gamma_{i,e}(m'|m)$. I assume that mortality shocks hit the household before marital transition shocks.¹⁵

Markets. Agents sell their time to a spot market for labor, receiving a fixed price of w . They can also save positive amounts of resources, i.e., they can rent assets at the market rate r .

Labor endowments. Agents are endowed with state dependent efficiency profiles, $\varepsilon_{i,m,e}$ for females and $\varepsilon_{i^*,m^*,e^*}$ for males. They also face idiosyncratic and persistent multiplicative income shocks (ϵ and ϵ^*). The processes generating these shocks are also state dependent. Hence, for males of age i^* , marital status m^* and education level e^* , labor income is given by

$$w\epsilon^*\varepsilon_{i^*,m^*,e^*}$$

Note that w is the market rental rate for efficiency units of labor. On the other hand, if children are present in the household, females need to devote some time taking care of them. These time requirements are reflected in $b(m,k) \in (0,1)$, so that labor income of females/mothers is given by

$$w\epsilon\varepsilon_{i,m,e}(1 - b(m,k))$$

¹³There is no child mortality risk nor multiple births

¹⁴This hazard rate is independent for each child in the household (if $k > 1$).

¹⁵This is important to calculate expectations over future states

Since I don't keep track of ages of children in the household, $b(m, k)$ is not time dependent. This simplifying assumption is in contrast of evidence that children require more time and money as they grow old.¹⁶

Preferences. Agents in the economy derive utility from per period consumption and the number of kids in the household. Hence, children are treated as durable goods in terms of utility and their characteristics (such as age and sex) are not qualities that enter agents utility function. In this paper I restrict attention to preferences that are separable in consumption and number of children of the form

$$u(c|z) + \gamma g(k)$$

Preferences for consumption depend on the characteristics of the household (z), namely, the number of members living under the same roof. This is to capture economies of scale in consumption and the idea that marriage might create consumption habits.¹⁷

Since the focus of this paper is on females and fertility, utility of married households is taken to be that of the female member. This could be the result of using unitary theories of the household or theories that allow for intra-household bargaining and the female having all the bargaining power. This assumption is restrictive, but necessary to keep this a feasible exercise.

Agents in this economy don't have the ability/desire of leaving bequests upon death and don't receive utility from their children once they leave the household.

The Dynamic problem when fertile. There are three distinct stages in the life-cycle of a female agent: (1) work-fertile stage, (2) work - infertile stage and (3) Retirement. Figure 4 presents the timing of events during the first stage, when females make both fertility and consumption/savings choices.

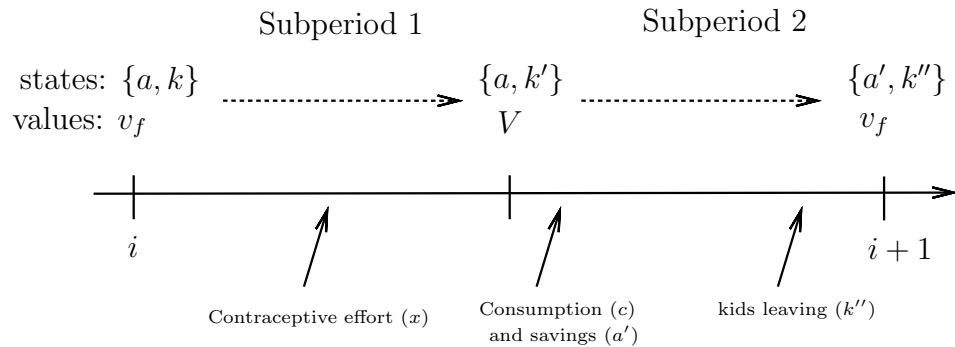


Figure 4: Time line of Events during fertile years

¹⁶For example, see Attanasio, Low, and Sánchez-Marcos (2008)

¹⁷See Hong and Ríos-Rull (2007).

As seen in the figure, agents in this stage transit between subperiod 1, where they make fertility decisions and subperiod 2, where they choose consumption and savings for the future. Before transiting to subperiod 1 again, households face an updating in their stock of children (due to kids leaving their mothers).

The following bellman equation represents the problem of agents during sub-period 2 (once they have made contraceptive effort choices):

$$\begin{aligned}
V(i, a, k, \tilde{z}) &= \max_{c, y} u(c|z) + \gamma g(k) + \delta_{i,e} \beta E [v_f(i+1, a', k', \tilde{z}')|z] \\
st : \\
c + y &= (1+r)a + w\epsilon_i \varepsilon_{i,m,e} (1 - b(m, k)) & \text{if } m = \{1, 3\} \\
c + y &= (1+r)a + w\epsilon_i \varepsilon_{i,2,e} (1 - b(2, k)) + w\epsilon_{i^*}^* \varepsilon_{i^*,2,e}^* & \text{if } m = 2, i^* < i_r \\
a' &= \Phi(y, z'|z)
\end{aligned} \tag{2}$$

where m represents current marital status ($m = 1, 2, 3$ stands for single, married and widowed/divorced respectively). The budget constraint accounts for different states, since married agents receive extra income from their spouses' labor, but only if the spouse is not retired ($i^* < i_r$). The Φ operator translates the amount of savings into next period assets given marital transitions and future states.¹⁸

Given optimal policies in subperiod 2, females make contraceptive effort choices in subperiod 1. The problem faced by them is:

$$\begin{aligned}
v_f(i, a, k, \tilde{z}) &= \max_x \pi(x|i, m, e) V(i, a, k, \tilde{z}) \\
&+ [1 - \pi(x|i, m, e)] \max \left\{ \begin{array}{l} V(i, a, k+1, \tilde{z}), \\ V(i, a, k, \tilde{z}) - \kappa_e \end{array} \right\} \\
&- C(x)
\end{aligned} \tag{4}$$

¹⁸The particular form of Φ is given by:

$$\Phi(y, z'|z) = \begin{cases} y & \text{if } (m' = 2|m = 2) \\ y & \text{if } (m' = 1, 3|m = 1, 3) \\ y & \text{if } (m' = 3|m = 2) \text{ (widowhood)} \\ \chi y & \text{if } (m' = 3|m = 2) \text{ (divorce)} \\ y + a^* & \text{if } (m' = 2|m = 1) \end{cases} \tag{3}$$

where (m', m) refers to a transition from m to m' next period. For example, when going from $m = 2$ (married) to $m = 3$ (through divorce), assets next period are a fraction χ of what is saved today, where $\chi \in (0, 1)$ reflects the partition of assets after a divorce. Note that when going from $m = 1$ (single) to $m = 2$ (married), assets next period are given by current savings plus what the prospective spouse brings to the household. This last variable (a^*) is a random variable that depends on the distribution of single agents of the opposite sex in the economy.

The value function at this stage is a convex combination of the continuation values with and without a new pregnancy. In the case of pregnancy (which occurs with probability $(1 - \pi(\cdot))$), agents have the chance of having an abortion at utility cost κ_e . Note that even though there are discrete outcomes following this stage (number of children in the household), the effort function convexifies the problem maintaining smoothness of the value function, which proves useful for solving (3) using standard continuous methods.¹⁹

My approach to model fertility choices differs from those who try to understand choices of specific birth control methods by women.²⁰ The setup above doesn't distinguish between different contraceptive methods nor their efficacy, but is general and its implementation straightforward.

Moreover, I allow the probability of no conception to be flexible enough so that overall fertility is not only due to failed birth control but also as the result of conscious efforts of females to start a family. Specifically, this means that the domain of π is the entire real line (contraceptive effort can be negative, in order to maximize the probability of conception) and the cost function is always positive, increasing away from zero. This general specification allows me to capture biological constraint on human fertility, which play a role in determining the optimal timing of births later in life.

The dynamic problem after fertile years. Once agents are past the fertile stage (cannot produce more children), they keep choosing optimal paths for consumption and savings until death. This stage in the life-cycle can also be divided into two: before and after retirement.

Before retirement ($i \leq i_r$), the problem of the agent is:

$$\begin{aligned}
 V(i, a, k, \tilde{z}) &= \max_{c, y} u(c|z) + \gamma g(k) + \delta_{i,e} \beta E [V(i+1, a', k', \tilde{z}')|z] & (5) \\
 \text{st :} & \\
 c + y &= (1+r)a + w\epsilon_i \varepsilon_{i,m,e} (1 - b(m, k)) & \text{if } m = \{1, 3\} \\
 c + y &= (1+r)a + w\epsilon_i \varepsilon_{i,2,e} (1 - b(2, k)) + w\epsilon_{i^*}^* \varepsilon_{i^*,2,e}^* & \text{if } m = 2, i^* < i_r \\
 a' &= \Phi(y, z'|z)
 \end{aligned}$$

The main difference between this Bellman equation and the one in (6) is that the the stock of children can only decrease from period to period.

After retirement, the problem reduces to

¹⁹Details of the numerical solution procedure are in the Appendix.

²⁰See for example Hotz and Miller (1993) and Rosenzweig and Schultz (1985).

$$\begin{aligned}
V(i, a, 0, \tilde{z}) &= \max_{c, y} u(c|z) + \gamma g(k=0) + \delta_{i,e} \beta E[V(i+1, a', 0, \tilde{z}')|z] \\
st : \\
c + y &= (1+r)a && \text{if } m = \{1, 3\} \\
c + y &= (1+r)a + w\epsilon_{i^*}^* \epsilon_{i^*, 2, e^*}^* && \text{if } m = 2, i^* < i_r \\
a' &= \Phi(y, z'|z)
\end{aligned} \tag{6}$$

at this stage no children are present in the household ($k = 0 \forall i \geq i_r$) and the only resources available for non-married agents are past savings. On the other hand, if agents are married to working age individuals, they enjoy the extra labor income $w\epsilon_{i^*}^* \epsilon_{i^*, 2, e^*}^*$.

4 Taking the Model to the Data

The solution of this model is a set of policy functions $x^{opt}(z|\Theta)$, $y^{opt}(z|\Theta)$ for contraceptive effort and savings respectively, given the current state z and other parameters, Θ (including prices). As it's usual, analytical expressions for the optimal policies are unfeasible, so I approximate them using numerical solutions to an empirical model with the following quantitative features.

Demographics and life-cycle. All agents start life at age 18 and cannot live longer than 95 years. Retirement is at 65 and the last fertile age is 40. A model period is one year when $i \in \{18, \dots, 40\}$, 5 years when $i \in \{40 + 1, \dots, 65\}$ and 10 years when $i \in \{65 + 1, \dots, 95\}$.²¹ Age specific mortality rates are taken from the National Center for Health Statistics and adjusted for educational attainment, as in Hong (2008).²²

I divide educational or skill types into those with at most a high school diploma or GED, and those with some post secondary education (college, community college, vocational school, etc.). To calculate the proportion of these types, I use the Current Population Survey (CPS) between 1990 and 1995. The proportion of high school individuals is around 40%. The majority of agents start life as single and childless, but I allow some of them to be married and have children. The proportion of never married 18 year old females in the CPS is around 93% and females with kids is around 9%. When performing simulations of the model, I distribute women uniformly according to these statistics to determine their initial state.

Since non-married females can always find a (new) partner in the model, I need information on who

²¹I do this to reduce the state space of the model. For details, see the Appendix.

²²Given the mapping from model periods to actual years, all age specific variables used in the computation are recalculated, depending on the stage of model life-cycle. Details in the Appendix.

they'd marry. Also from the CPS, I compute the proportion of couples by age and educational attainment of the partners, the age distribution of male partners for married females and the relative asset position of both non-married males and non-married females.²³ Given this information, I construct education-specific grids with probabilities of marrying someone of characteristics given by $\{e^*, i^*, a^*\}$ (education, age and assets of prospective husbands). Since I'm not computing equilibrium, this procedure doesn't check for internal consistency of measures of agents (as in Hong and Ríos-Rull (2007), where all these probabilities are endogenous objects).

Transitions between marital states come from the Panel Study of Income Dynamics (PSID) for the years 1990-1995. I follow all heads of household older than 18 years old (inclusive) and compute annual age and education specific transition probabilities between three states: single, married and divorced/widowed. Given variable specification in the PSID, married couples include cohabitating couples.

Preferences. I use an additively separable specification for instantaneous, per period utility: $u(c|z) + \gamma g(k)$. The marginal utility from consumption depends on the size of a household:

$$u(c|z) \equiv \tilde{u} \left(\frac{c}{1 + \mathbf{1}_{\{m=2\}} \phi_m + \mathbf{1}_{\{k>0\}} k \phi_k} \right) \quad (7)$$

Where $\mathbf{1}_{\{cond\}}$ is the indicator function that takes a value of one when "cond" is true and zero otherwise; ϕ_m and ϕ_k are equivalence scales which discount consumption in married households and households with children respectively. If $\phi_m, \phi_k < 1$, economies of scale in consumption exist in the household: expenditures to maintain the level of per capita utility constant, grow proportionally less than the number of household members.

The specific functional forms for $\tilde{u}$ and g are given by

$$\tilde{u}(\mathbf{c}) = \frac{\mathbf{c}^{1-\eta_c} - 1}{1 - \eta_c} \quad g(k) = \frac{(1+k)^{1-\eta_k} - 1}{1 - \eta_k} \quad (8)$$

Fertility. I use the following function for π (the probability of NO conception, given effort x):

$$\pi(x|i, m, e) = \begin{cases} \pi^+(x|i, m, e) & \text{if } x > 0 \\ \pi^-(x|i, m, e) & \text{if } x < 0 \end{cases} \quad (9)$$

where

$$\pi^+(x|i, m, e) = \frac{\exp\{x\}}{\exp\{x\} + \varphi_{i,m,e}^+ \exp\{-x\}}$$

²³My proxy for individual assets is the sum of interest, dividend and rent income as defined in the March supplements of the CPS

and

$$\pi^-(x|i, m, e) = \frac{\exp\{x\}}{\exp\{x\} + \varphi_i^- \exp\{-x\}}$$

In general, π is a modified logistic function with φ as a shift parameter. Regalia and Ríos-Rull (2001) use a similar framework to study fertility choice in an equilibrium model. Note that the higher $\varphi_{i,m,e}^+$, the higher the probability of a pregnancy when effort (x) is positive (females trying to avoid fertility), which means that I can parameterize higher difficulty in controlling fertility by increasing $\varphi_{i,m,e}^+$. However, if women are trying to get pregnant (negative x), parameterizing π through the same $\varphi_{i,m,e}^+$ would not be realistic, since it would mean that ability in using contraceptive methods is negatively correlated with the ability of procuring a conception when it is desired. Hence, I use a different shifter, φ_i^- for this case. Note that contraceptive ability $\varphi_{i,m,e}^+$ depends on age, marital status and education and its parameterization is given by

$$\varphi_{i,m,e}^+ = \tilde{\varphi}_{i,m}^+ + \mathbf{1}_{\{e=\underline{e}\}} \bar{\varphi}_i$$

While ability in using contraceptive technology might depend on marital status and education, I assume that the technology of procuring a pregnancy depends mostly on biological constraints. This is represented by φ_i^- depending only on age.

I parameterize these age profiles using polynomial approximations on age: given the order for the polynomial (p , the same for all profiles) the number of parameters to be determined is then $p + 1$ (the number of polynomial coefficients) times 5 profiles: for singles ($\tilde{\varphi}_{i,1}^+$), for married ($\tilde{\varphi}_{i,2}^+$), for divorced/widowed ($\tilde{\varphi}_{i,3}^+$), for the extra risk faced by the high school group ($\bar{\varphi}_i$) and for the biological fertility profile φ_i^- . For example, the profile for married individuals is given by

$$\tilde{\varphi}_{i,2}^+ = \max\{\alpha_0^2 + \alpha_1^2 i + \alpha_2^2 i^2 + \dots + \alpha_p^2 i^p, 0\}$$

while the profile for the excess fertility risk faced by the high school group is²⁴

$$\bar{\varphi}_i = \max\{\bar{\alpha}_0 + \bar{\alpha}_1 i + \bar{\alpha}_2 i^2 + \dots + \bar{\alpha}_p i^p, 0\}$$

I reduce the number of parameters by assuming that the fertility control technology for singles is the same than for divorced/widowed agents ($\tilde{\varphi}_{i,1}^+ = \tilde{\varphi}_{i,3}^+$).

On the other hand, I parameterize the utility cost of exerting contraceptive effort as

$$C(x) = \frac{x^2}{2} \xi$$

²⁴The max operator ensures that the resulting shifter parameter is non negative, restriction I impose to get well behaved profiles when they are decreasing (they may become explosively negative at higher ages)

This cost function is symmetric around zero, so I use it for both sides of the fertility problem: when females want to prevent or are seeking a pregnancy. This is not restrictive, given the asymmetric structure of π .

Earnings and Labor Supply. Endowments of labor efficiency profiles come from the CPS (years 1990-1995). I calculate annual labor earnings for the two educational groups (high school and college), by age and marital status. As in Hong and Ríos-Rull (2007) and Hong (2008), I use annual earnings since they capture differences in the intensive margin of earnings by sex and marital status better than hourly earnings. To account for inflation, I adjust nominal values by the GDP deflator for the year 2000.

I restrict attention to childless females throughout the sample period. For males, I don't make that distinction, since the change in income due to the presence of own children in the household is not significant.

I attribute the time cost of child-rearing $b(m, k)$ to annual labor income differentials of females in fertile age (18 to 40) by number of children. This is different than accounting for hours worked by number of children in the household; it stands alternatively for different ways in which a child might change earnings ability of the mother (e.g., getting a job with more flexible schedule but lower pay, getting a job with lower pay but closer to home, not getting tenured at an academic job or not being made partner at the firm, etc.) other than through hours worked per week. The computed values are in table 1.

Children	$b(m = \{1, 3\}, k)$	$b(m = 2, k)$
0	0%	0%
1	5.9%	26.5%
2	16.9%	37.5%
3	41.0%	52.6%
4	61.3%	63.3%
5+	81.2%	72.8%

Table 1: Time cost of Children (in terms of full time work), CPS 1990-1995.

As seen from the table, time cost of children (or time away from the best paid market alternative) is increasing in the number of children present in the household. Note also that the cost increases faster in the number of kids for married women than for single ones.

For earnings shocks, I use an AR(1) specification

$$\epsilon'_e = \rho_e \epsilon_e + \mu'_e \quad (10)$$

where $\mu_e \sim N(0, \sigma_e)$. These shocks are gender and education specific. I take values of ρ_e, σ_e (for $e = \{\underline{e}, \bar{e}\}$) from Hong (2008), who uses the PSID between 1986-1992 to compute maximum likelihood

estimates. As is common, I discretize both continuous processes using the method proposed by Tauchen (1986).

5 Estimation

Given the partial equilibrium nature of the exercise, I set several model parameters exogenously. First, the rental price of efficiency units of labor w is normalized to 1. I set the interest rate to equal the average of the 1-year Treasury Bill Rate (monthly auction averages).²⁵ I let the discount factor β to be $1/(1+r)$. For equivalence scales, I use $\phi_m = 0.7$ and $\phi_k = 0.5$ (i.e., the OECD values).

The rest of the model parameters are determined jointly, by minimizing the square difference between data and model moments. The procedure is standard in the literature: (i) select which data targets to use (ii) guess values for model parameters (iii) solve the model and calculate optimal policies (iv) simulate life-cycles for a large number of individuals and compute model equivalents to the data targets (vi) calculate the error of the iteration (the sum of square values of the difference between every data and model moment) (vii) if the error is less than a pre-specified tolerance, exit; if not, update parameters according to some predefined rule and repeat from step (iii) until convergence. This is a simplified simulated method of moments estimation procedure, where the weighting matrix for moments is the identity matrix.

The list of moments is as follows:

- Age profile of pregnancy rates for non-married females by education²⁶: 46 moments = 23 ages $\times$ 2 education levels
- Age profile of pregnancy rates for married females by education: 46 moments = 23 ages $\times$ 2 education levels
- Age profile of abortion rates by education: 46 moments = 23 ages $\times$ 2 education levels
- Age profile of unplanned pregnancy rates by education: 46 moments = 23 ages $\times$ 2 education levels

In total, there are 184 moments to match. On the other hand, the number of model parameters depends on the choice of order for the polynomials that define age profiles for fertility parameters. I chose $p = 6$, so each age profile is defined by 7 coefficients. Thus, the model has 34 parameters to be determined jointly:

- curvature in the utility of consumption η_c (1)

²⁵Series id TB1YA, on the St. Louis Fed Economic Data webpage.

²⁶Note that I merge the statistics for both single and widowed/divorced females.

- curvature in the utility of children η_k (1)
- multiplicative parameter in utility of children γ (1)
- utility cost of an abortion κ_e (2)
- utility cost of contraceptive effort ξ (1)
- contraceptive ability parameter $\tilde{\varphi}_{i,m}$ (14 parameters = 7 coefficients $\times$ 2 marital states)
- conception ability parameter ϕ_i^- (7)
- contraceptive ability shifter for low skill/education group $\bar{\varphi}_i$ (7)

Solution to the model is by backwards recursion. In the last period of life there is no continuation value (I assume no bequests motives nor life insurance) hence optimal policies and value functions can be calculated recursively from the next to last period. Details of the procedure are in the Appendix.

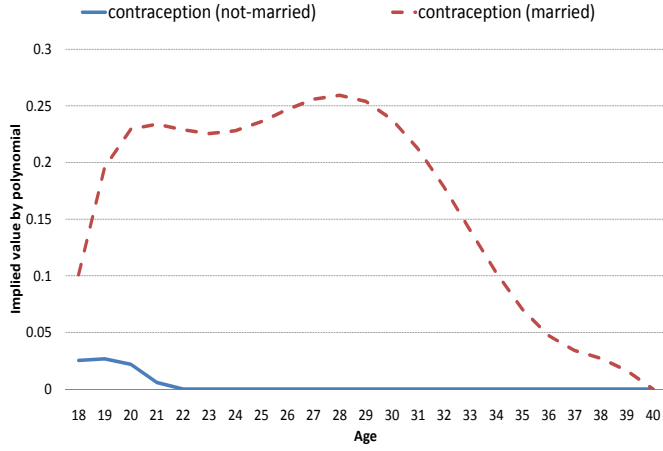
6 Results

The estimated parameters are in table 2. The full list of coefficients for the polynomial functions used in parameterizing conception probabilities is in the Appendix. In figure 5, I present instead the implied age profiles given by the estimated polynomials.

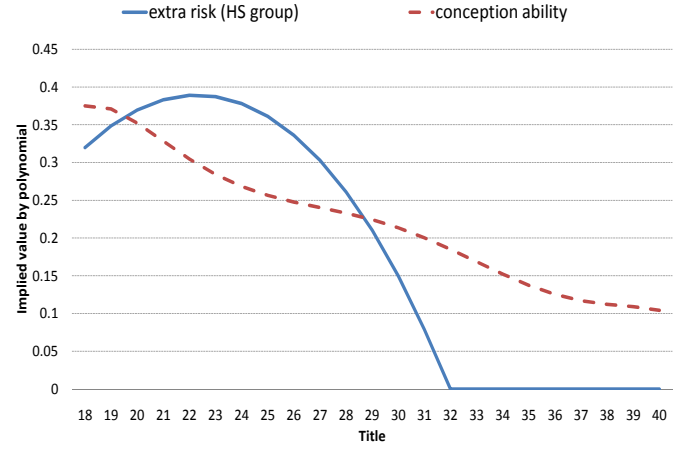
Parameter	value
η_c	1.74
η_k	1.82
γ	0.82
κ_{HC}	5.13
$\kappa_{College}$	1.21
ξ	1.58

Table 2: Model Parameters

The estimated curvature in the utility of consumption (η_c) equals 1.74, which is in line with the rest of the literature (the usual number lays between 1.5 and 2). Preferences are close to being homothetic: the value of η_k (1.82) is close to the one for η_c . Overall, the value of both parameters indicate that consumption and children enter as complements in the utility function, so females enjoy the presence of children more when consumption levels are higher. Given increasing wages during the earlier stages of the life-cycle, this means that females would like to postpone childbearing as much as possible.



(a) $\tilde{\varphi}_{i,\{1,3\}}$ and $\tilde{\varphi}_{i,2}$



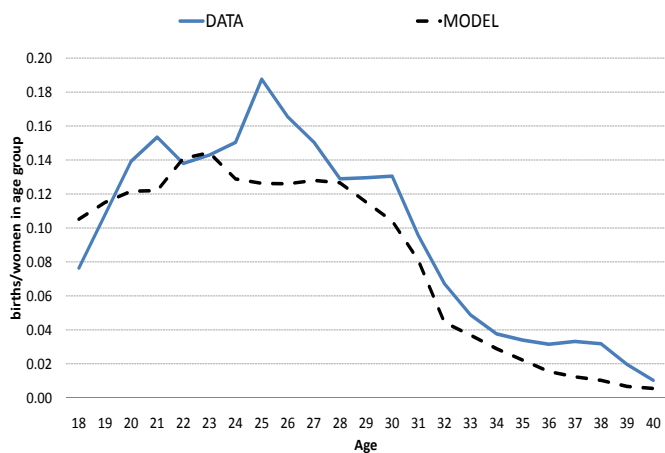
(b) $\bar{\varphi}_i$ and ϕ_i^-

Figure 5: Estimated parameters for contraceptive technology and conception ability. This figure shows the implied age-specific values given the model-estimated polynomials

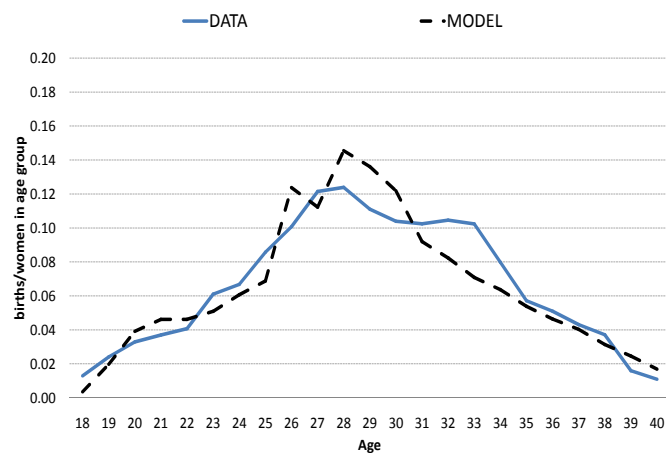
The utility cost of an abortion is higher for high school individuals than for college individuals (5.13 vs. 1.21 respectively). Since this is a utility cost, it captures both monetary and psychological disincentives to get an abortion. The higher parameter value for the high school group is a reflection of the evidence in figure 3, where the fraction of unwanted births that end up being aborted is lower for that educational group.

Figure 5 shows the age-profiles that describe fertility technology and restrictions during the life-cycle. All profiles are decreasing in age which is a reflection of decreasing chances of conception late in the fertile stage of life. Contraceptive parameters for married individuals are higher than for single ones, which means that birth control is easier when there is no steady partner of the opposite sex in the household. With respect to the risk faced by the high school group, results imply that females in that group have 14% more chances (on average in their lifetime) of having an unwanted pregnancy than their college counterparts, given the same amount of contraceptive efforts. The calculation of this percentage comes from using the age profile $\bar{\varphi}_i$: the approximated probability of an unwanted pregnancy (when exerting a very low amount of effort) for every age is given by $\bar{\varphi}_i/(1 + \bar{\varphi}_i)$ (The average for all ages is 13.98%).

Figures 6 to 8 show the goodness of fit of the model. Overall, the model does a good job in replicating the stylized facts with respect to the number and timing of births across educational groups.

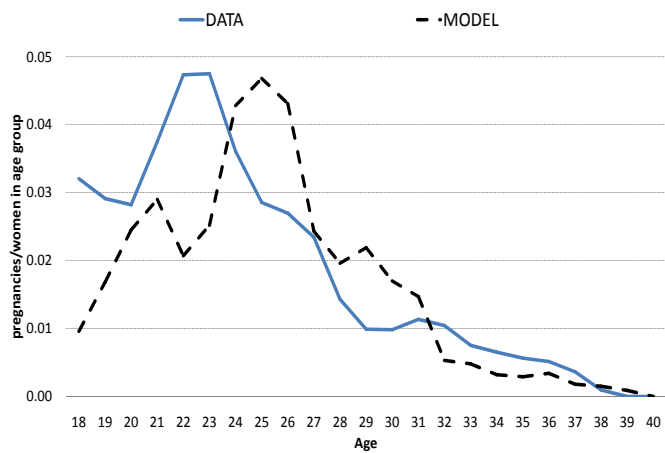


(a) High School

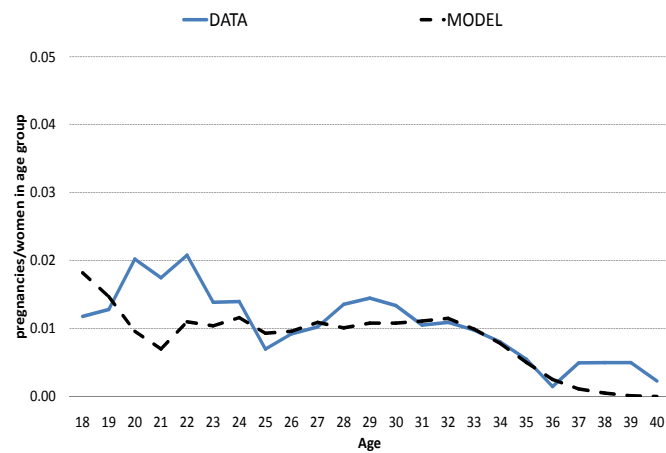


(b) College

Figure 6: Age specific fertility rates: Data and Model

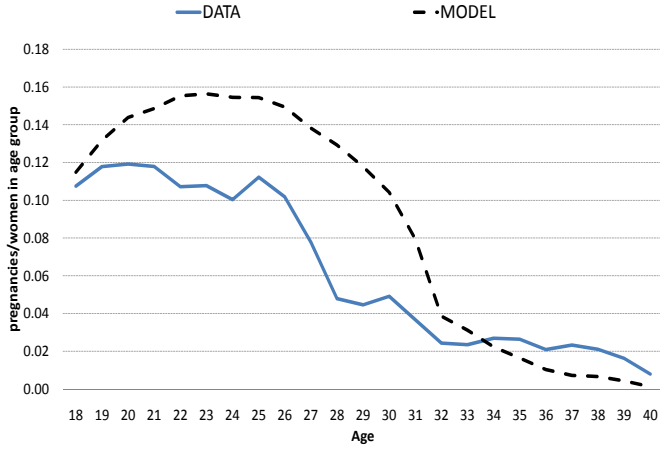


(a) High School

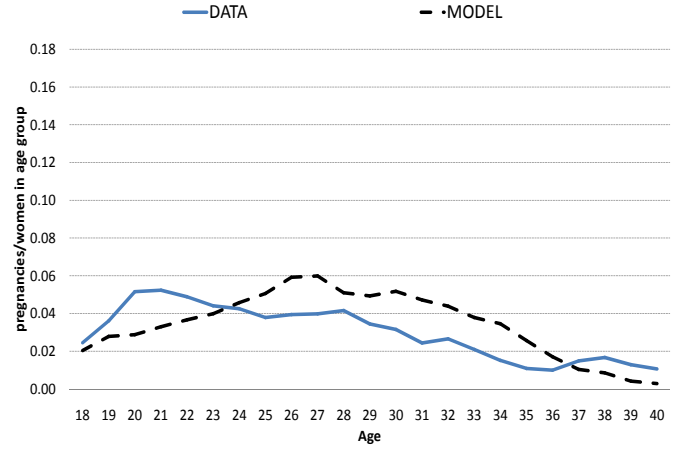


(b) College

Figure 7: Abortion rates: Data and Model



(a) High School



(b) College

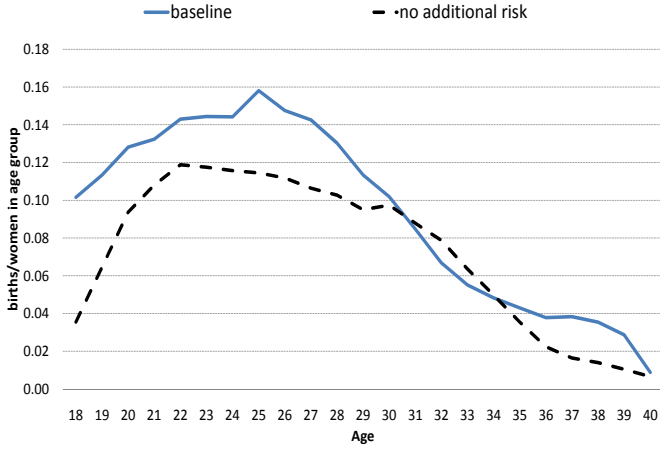
Figure 8: Unwanted rates: Data and Model

Both simulated abortion and fertility rates follow closely their data counterparts; on the other hand, the rate of unwanted pregnancies is overpredicted for the high school group but the overall qualitative features of the data are preserved (differences among educational groups and timing of pregnancies).

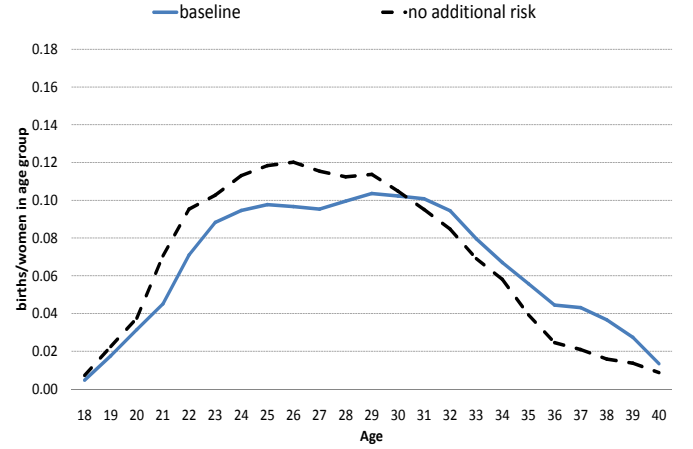
Relevant Margins Below I present exercises that show the importance of the two main ingredients of my theory, namely differential fertility risk across educational groups and self-insurance (the ability to save for the future).

Figure 9 shows the comparison between the baseline model and the case when the model is calibrated to match the facts without extra risk for the high school group ($\bar{\varphi}_i = 0, \forall i$). As seen from the figure, the latter model is unable to match the higher fertility rates of high school educated individuals, while the rates for the college group are slightly overpredicted. In terms of timing of births, this alternative model correctly accounts for the delay in childbearing by females in the college group as opposed to those in the high school group; however, this difference is less pronounced than in the baseline model.

In the model where the two educational groups face the same contraceptive technology, the ratio between their TFRs is 0.94 (1.56 and 1.67 for high school and college educated individuals respectively) while the ratio in the data (and the baseline) is closer to 0.7. Hence, differential fertility risk earlier in life accounts for most of the differences in the number of births across educational groups during the life-cycle, leaving a small role for differences in wages. This result hints that differences in wage profiles



(a) High School



(b) College

Figure 9: Age-specific fertility rates, different models

help mostly in predicting the different timing of births: the flatter profiles of life-cycle wages of high school educated females makes them choose early childbirth (at the margin) given the complementarity between consumption and children.²⁷ On the other hand, college individuals face rapidly growing wage profiles, which induce them to delay fertility.

In the next simulation, I take the baseline model, shut down the ability to save and recalibrate the economy. Since I assume that after retirement agents don't receive any income other than past savings, I set the last period in the no-savings simulation to be i_r and discard the retirement stage (setting the value functions after age i_r equal to zero for any point in the state space).

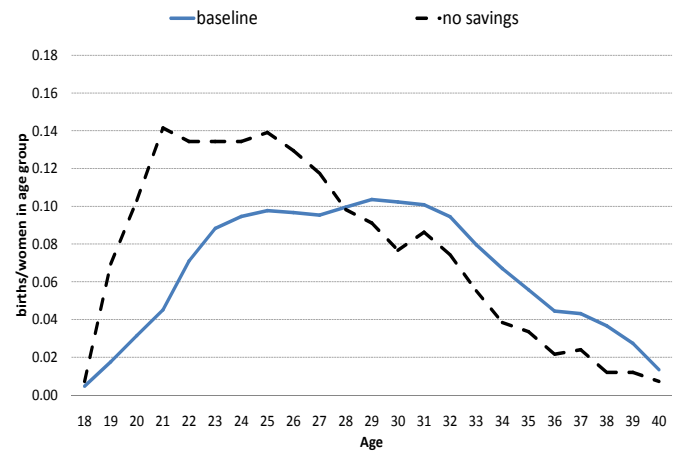
The predicted fertility profiles of the no-savings case are in figure 10. Results from simulating this version of the model (no savings but with differential fertility risks) shows that the choice of assumption regarding capital markets changes the predictions of fertility models in non-trivial directions. It also helps in understand the intertemporal margins faced by individuals in the life-cycle.

Given the inability to save, all income has to be consumed at the end of each period. For individuals in both educational groups, consumption in this setting is, on average, higher than what they would have chosen had they been able to save. For individuals in the high school group, this leads to marginal lower fertility rates than in the baseline. The college group, however, faces increasing wage profiles,

²⁷After re-estimating the parameters, $\eta_c = 1.17$ and $\eta_k = 1.11$, so consumption and children are still complements, but the extent of this complementarity is smaller than in the baseline



(a) High School



(b) College

Figure 10: Age-specific fertility rates, different models

thus increasing utility. Given complementarity between consumption and children in the utility function, this 'forced' higher consumption for the group leads them to conceive more children to increase utility when wages and consumption are high. Since fertility choice is dynamic and females are restricted to one child per period, the timing of first child birth is shifted towards younger ages, which is a major counterfactual prediction of this variation of the model.

7 Conclusion

In this paper I study life-cycle fertility in the U.S., focusing on birth profile differences across educational groups (high school and college). To understand the facts on timing, number of births and abortions during the life-cycle, I develop a structural model where agents transit through different marital states, face idiosyncratic survival and earnings risk and capital markets are incomplete (individuals cannot borrow against their future earnings). In this setting, I include endogenous decisions about fertility, but where the outcome of those decisions cannot be controlled perfectly (i.e., fertility risk).

Using data from the National Survey of Family Growth, the Panel Study of Income Dynamics and the Current Population Survey, I estimate the model using the simulated method of moments. From the exercise, I conclude that differential fertility risk (in the form of ability to control fertility plans) across education groups is the main determinant of differences in timing and levels of fertility. The main identification mechanism for the parameters in the model is the differential timing in the rates of unwanted pregnancies versus abortions for the different educational groups.

My results show that standard fertility theories, which rely solely on substitution effects to produce negative skill-fertility relationships, cannot account for life-cycle nor cross sectional facts. Furthermore, I show that fertility risk is an important source of life-cycle inequality and that life-cycle models built to ask questions about economic inequality in environments without complete insurance should incorporate endogenous and risky fertility decisions.

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8 Appendix

8.1 Data

Figure 11 shows the profiles for labor endowments, computed from march supplements of the Current Population Survey (years 1990 to 1995). In the figure I show annual earnings for females, between ages 18 to 65, corrected for inflation using the GDP deflator for the year 2000. These profiles are smoothed using a 5th order polynomial.

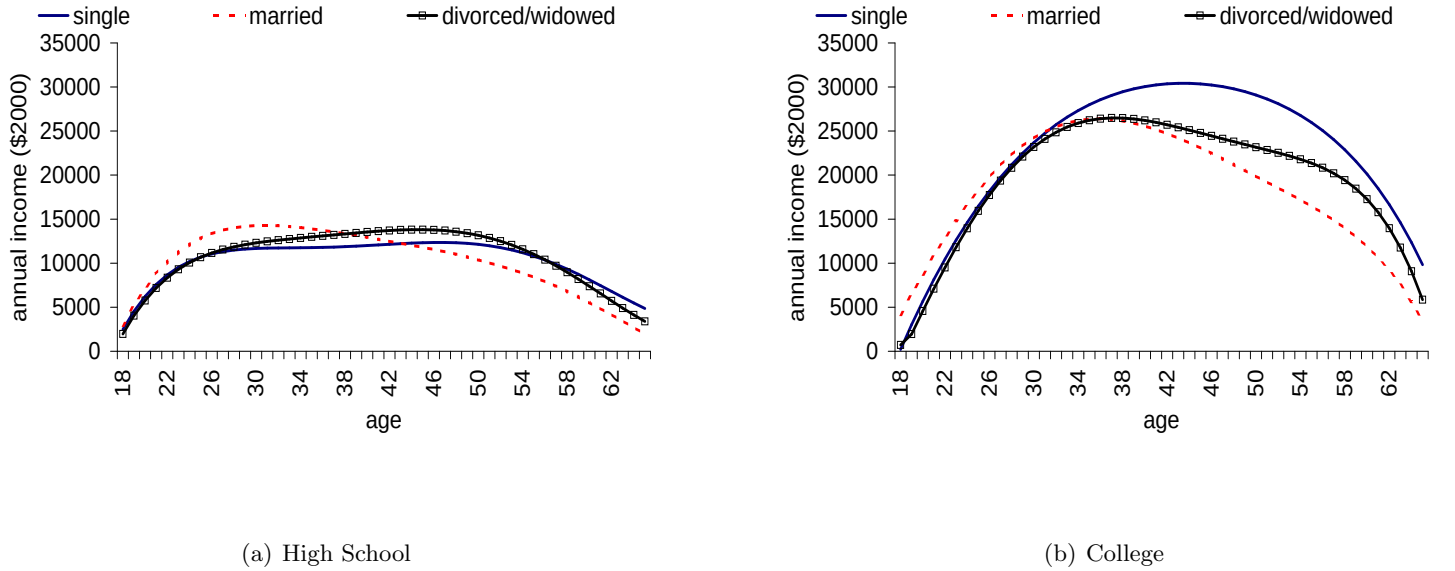


Figure 11: Labor Endowments by educational group

To characterize the labor market, I also use gender and education specific idiosyncratic labor shocks. These shocks come from estimates from Hong (2008), who uses labor earnings data from the PSID to calculate the unobserved component of annual labor earnings. I use a standard discretization of the continuous AR(1) described in the paper. I choose to discretize the four processes (2 education groups and 2 genders) by a 3 state markov system. The standard in the literature is to use at least 5 states, but computational burden prevents me from using a more detailed shock structure. However, results in the paper don't rely in the dimensionality of these shocks.

Also from the CPS, I calculate the proportion of females (by education) married to college educated males (irrespective of presence of children in the household), in order to measure positive assortative matching in the marriage market. As seen in figure 12, marriage indeed shows the positive assortative matching property.

I compute yearly survival probabilities by educational group using the information in Hong (2008). I interpolate his 5 year values and smooth the resulting series with a second order polynomial. The resulting probabilities for female individuals are in figure 13.

To calculate transitions through marital states, I use the Panel Study of Income Dynamics (PSID) for

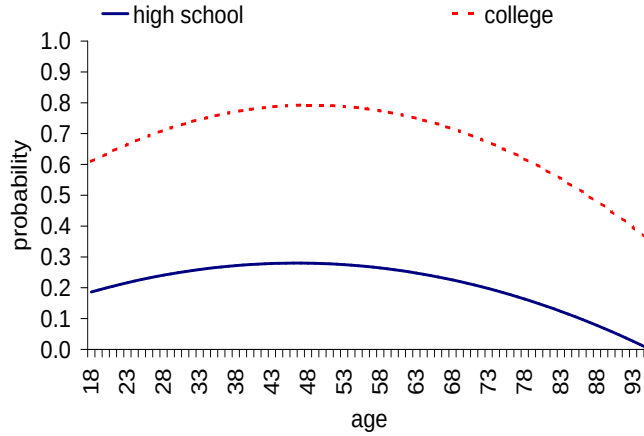


Figure 12: Probability of being married to college male, by education of female

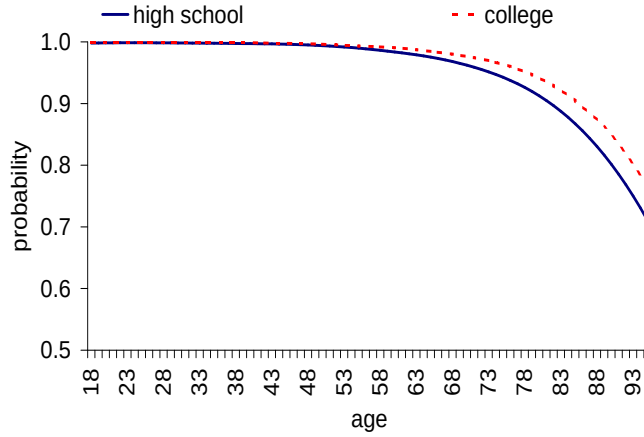
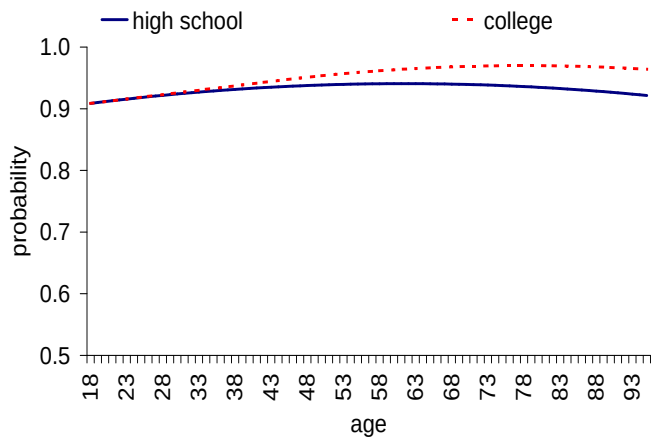


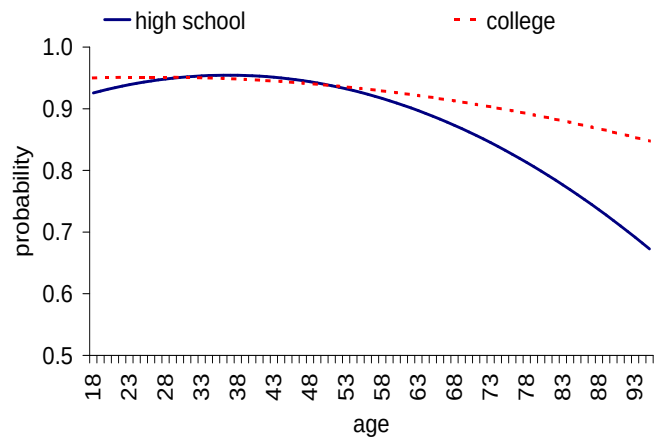
Figure 13: Survival probability by education

the years 1990 through 1995. I use heads of household and wives (as defined in the PSID) to compute the following probabilities, by education and age: probability of remaining single, the probability of remaining married and the probability of getting married conditional on being divorced/widowed. Given these three probabilities, I can span all transitions (e.g., some probabilities are zero by definition and others are just complements). I extrapolate these probabilities when necessary since the PSID doesn't have many observations for young/old heads of household. Given the short span of my chosen sample, individuals contribute at most 5 observations/years, making these probabilities a cross-section description of marital transitions during the mid 1990s in the U.S. Figure 14 shows these transitions.

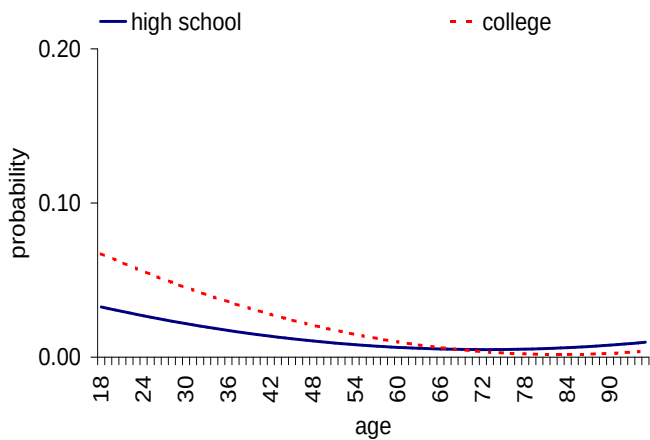
I assume simple age and asset distribution of prospective male partners. For ages I consider only 3 possible alternatives: same age, one year older and two years older ($i^* = \{i, i+1, i+2\}$), each occurring with probabilities $P(i^* = i) = 0.4$, $P(i^* = i+1) = 0.41$ and $P(i^* = i+2) = 0.19$, which come from CPS data. Age of partners is important since they determine the extra income for the household in terms of partner's labor earnings and the probability of death (hence, transitioning to widowhood status). Since the profiles for both characteristics are smoothed, the tradeoff between accuracy and simplicity of



(a) single to single



(b) married to married



(c) div/wid to married

Figure 14: Transition probabilities for marital states

the solution by assuming such a narrow age distribution is lessened.

For assets, I calculate from CPS data the average annual non-labor income (dividends, interests and rents) for both single males and females. Single males have on average 20% higher non-labor income than single females. Hence, I create a simple three point distribution for assets of prospective partners $a^* = \{1.1a, 1.2a, 1.3a\}$, centered around the fact that on average $a^*/a = 1.2$. This simple distribution is uniform (equal probabilities for each point). Changing this distribution doesn't alter any of the qualitative results from the exercise.

A note on Total Fertility Rates and cohort effects: throughout the paper, I assume no cohort effects in fertility rates. Although fertility has experienced significant changes during the 20th century, fertility rates for the cohorts considered in my analysis are quite stable. Figure 15 shows age specific fertility rates computed from the internet release of Vital Statistics of the United States for the year 1995 (tables 1-7).

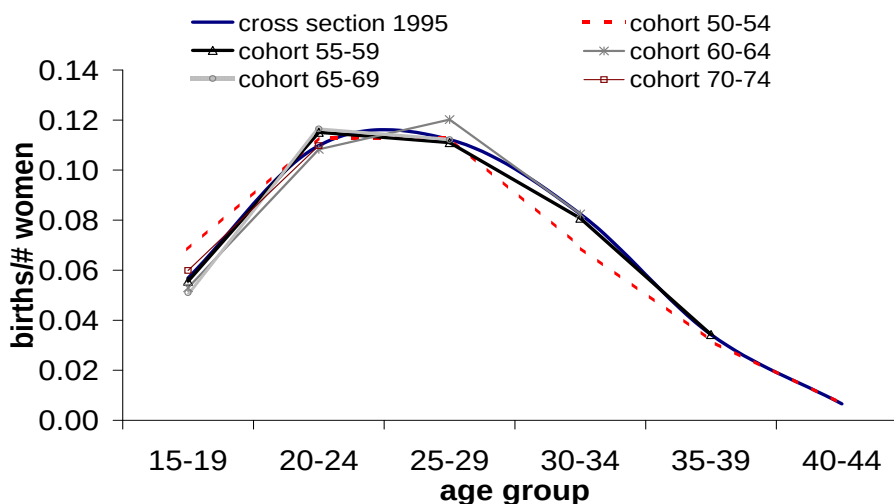


Figure 15: Age Specific Fertility Rates by birth cohort

The figure shows both total fertility rates for the cross section in 1995 and for cohorts (denoted by year of birth) across multiple survey years. The differences between the cross-sectional profile and actual cohort profiles is minimal. This comes from the fact that I am considering a small window in the life-cycle of cohorts that are close together (at most 20 years between births).

8.2 Computation and Estimation details

To solve the model, I use a Chebyshev regression (as described in Judd (1998) and Heer and Mauner (2004)) to approximate the optimal policies for savings and contraceptive effort and the value function along the asset space (the only continuous state variable in the model). My approximation is described by 7 collocation points and the use of a Chebyshev polynomial of degree 5. Increasing both the number of collocation points and/or the order of the polynomial doesn't improve the quality of the approximation significantly.

A note on the non-standard mapping between model periods and years: I adopt this procedure to save on computational time. A similar feature is present in Keane and Wolpin (2006). In my model, $i_f = 40$ (last fertile age), $i_r = 45$ (stands for a retirement age of 65 years) and $I = 48$ (represents the age of 95, the last period of life).

I assume that an individual aged $i \in \{i_f + 1, \dots, i_r\}$ experiences one model period as the average of 5 real years; when $i \in \{i_r + 1, \dots, I\}$, the experience is that of 10 averaged years. The external data used (and described in the previous section) is treated accordingly depending on the age of the individual: earnings, survival and transition probabilities, etc., are averaged in groups of 5 or 10 years accordingly.

Some notes on the estimation: The parameterization of age profiles for contraceptive and conception ability are in table 3

	$\tilde{\varphi}_{i,m=\{1,3\}}^+$ (not-married)	$\tilde{\varphi}_{i,2}^+$ (married)	$\bar{\varphi}_i$ (high-school)	φ_i^- (conception)
constant	1.8953E-02	-1.0200E-01	1.0738E-01	3.5130E-01
i	9.0301E-03	2.7700E-01	4.6700E-02	4.3360E-02
i^2	-2.8517E-03	-8.5670E-02	-5.7464E-03	-2.2970E-02
i^3	2.0009E-04	1.2520E-02	1.5478E-04	3.5760E-03
i^4	-4.2137E-06	-9.1220E-04	-8.5060E-09	-2.5940E-04
i^5	-1.3230E-05	3.1810E-05	-3.7080E-09	8.8510E-06
i^6	-4.4720E-07	-4.2400E-07	-3.1730E-09	-1.1470E-07

Table 3: polynomial coefficients for parameterization of fertility profiles

To accelerate the estimation algorithm, I use a Beowulf cluster with 20 processors. I parallelize at the parameter level, using the APPSPACK software available free on the web. See Gray and Kolda (2006) and Kolda (2005) for details.